

Control work 7, and work on the mistakes

Trigonometric equations and inequalities in one paper, then a named diagnosis of each slip.

LESSON 89–90

2 × 40 MIN

ALGEBRA 10, NAZORAT ISHI 7

P1 5 REVIEW

LESSON CLOCK

3

42'

12'

25'

8'

Instructions	3 min
The paper	42 min
Answers	12 min
Diagnosis and rewrite	25 min
Targets	8 min

LEARNING OBJECTIVES

- Solve the four basic equations and their standard extensions under time.
- Select solutions in a given interval without losing a family.
- Solve an inequality and write the answer as intervals.
- Name each lost mark and rewrite the solution in full.

TEXTBOOK

National	pp. 301–304
Cambridge	Pure Mathematics 1, pp. 121–124
Workbook	Control paper A7

01

Explanation

The paper — 35 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Solve $\sin x = -\frac{\sqrt{3}}{2}$ in general and on $[0, 2\pi]$	5	L79–81
2	Solve $\cos 2x = \frac{1}{2}$ on $[0^\circ, 360^\circ]$	5	L79–81
3	Solve $2\sin^2x + 3\sin x - 2 = 0$, rejecting the impossible root	6	L82–85
4	Solve $\sin 2x = \sqrt{3}\cos x$ by factorising	6	L82–85
5	Write $\sin x + \sqrt{3}\cos x$ as $R\sin(x + \alpha)$, then solve $= 1$	7	L82–85
6	Solve $\cos x < \frac{1}{2}$ and give the answer on $[0, 2\pi]$	6	L86–88

WHERE THE MARKS ACTUALLY GO

Q1 and Q2 each carry one mark for the **second** family, Q3 one for rejecting $t = -2$, Q4 one for factorising rather than dividing, Q6 one for reading the arc anticlockwise. Five of the thirty-five marks are for not losing something.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
lost family	one answer where two were wanted	draw the chord and count the points
divided by $\cos x$	$\sin 2x = \sqrt{3} \cos x \Rightarrow 2 \sin x = \sqrt{3}$	factor; keep $\cos x = 0$
root not rejected	$\sin x = -2$	check $ t \leq 1$ before solving
wrong period	$\tan x = a \Rightarrow \dots + 2\pi n$	tangent steps by π
arc read backwards	an empty interval	anticlockwise, always
substitution not undone	answer given in t or in $2x$	divide by the coefficient at the end
$\arccos(-a)$ mishandled	$-\arccos a$	$\pi - \arccos a$

Write the slip's name in the margin of every question you lost a mark on, then rewrite the whole solution — not the wrong line.

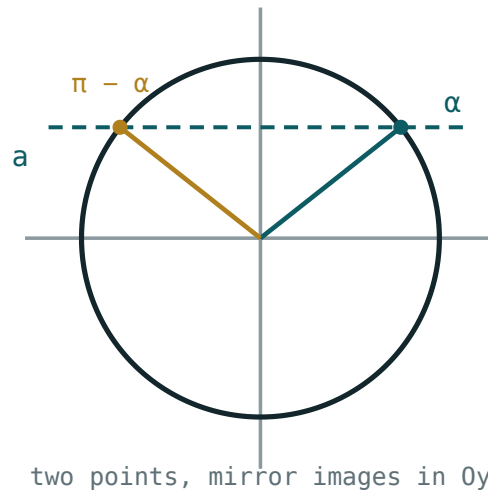
The quarter so far, in one page

EVERYTHING OF LESSONS 79–88

$$\sin x = a \Rightarrow x = (-1)^n \arcsin a + \pi n \quad \cos x = a \Rightarrow x = \pm \arccos a + 2\pi n$$

$$\tan x = a \Rightarrow x = \arctan a + \pi n \quad a \sin x + b \cos x = R \sin(x + \alpha), R = \sqrt{a^2 + b^2}$$

Three moves for the rest: **substitute**, **factorise**, **reduce**. Inequalities: solve the equation, shade the arc, read it anticlockwise.



One picture that generates every formula above.

Lessons 91–98 leave trigonometry entirely for probability. Nothing of this block is needed there — but all of it returns in Grade 11, where the derivative of $\sin x$ makes these equations the last step of an optimisation rather than the whole question.

02 Worked examples

EXAMPLE 1 Model answer, Q2: solve $\cos 2x = \frac{1}{2}$ on $[0^\circ, 360^\circ]$.

- ① Let $t = 2x$, so $t \in [0^\circ, 720^\circ]$.
Widen the interval first.

- ② $t = \pm 60^\circ + 360^\circ n \Rightarrow t = 60^\circ, 300^\circ, 420^\circ, 660^\circ$
Four values in range.

- ③ $x = 30^\circ, 150^\circ, 210^\circ, 330^\circ$
Halve each.

ANSWER

$30^\circ, 150^\circ, 210^\circ, 330^\circ$

EXAMPLE 2 Model answer, Q4: solve $\sin 2x = \sqrt{3} \cos x$.

$$(1) \quad 2 \sin x \cos x - \sqrt{3} \cos x = 0$$

$$(2) \quad \cos x (2 \sin x - \sqrt{3}) = 0$$

Factor, do not divide.

$$(3) \quad x = \frac{\pi}{2} + \pi n$$

$$(4) \quad \sin x = \frac{\sqrt{3}}{2} \Rightarrow x = (-1)^n \frac{\pi}{3} + \pi n$$

ANSWER

$$\frac{\pi}{2} + \pi n \text{ and } (-1)^n \frac{\pi}{3} + \pi n$$

EXAMPLE 3 Model answer, Q5: write $\sin x + \sqrt{3} \cos x$ as $R \sin(x + \alpha)$ and solve $= 1$.

$$(1) \quad R = \sqrt{1+3} = 2, \tan \alpha = \sqrt{3} \Rightarrow \alpha = \frac{\pi}{3}$$

$$(2) \quad 2 \sin(x + \frac{\pi}{3}) = 1$$

$$(3) \quad x + \frac{\pi}{3} = \frac{\pi}{6} + 2\pi n \text{ or } \frac{5\pi}{6} + 2\pi n$$

$$(4) \quad x = -\frac{\pi}{6} + 2\pi n \text{ or } x = \frac{\pi}{2} + 2\pi n$$

ANSWER

$$x = -\frac{\pi}{6} + 2\pi n, x = \frac{\pi}{2} + 2\pi n$$

03 Interactive model

Work Q4 twice on the board — once by factorising, once by dividing — and let the class name the lost family.

Ten questions on the trigonometry block

$\sin x = 1.5$ has:

two solutions

one

none

infinitely many

$\cos x = a$ steps by:

π

2π

$\frac{\pi}{2}$

4π

$\tan x = a$ steps by:

π

2π

$\frac{\pi}{2}$

3π

$\arccos(-\frac{\sqrt{2}}{2})$ is:

$$-\frac{\pi}{4}$$

$$\frac{3\pi}{4}$$

$$\frac{\pi}{4}$$

$$\frac{5\pi}{4}$$

To solve $2\cos^2x + \sin x = 2$ you first:

divide by $\cos x$

replace $\cos^2x$ by $1 - \sin^2x$

square

substitute $t = 2x$

$\sin x \cos x = \sin x$ should be:

divided

factorised

squared

left alone

A homogeneous equation is divided by:

$\sin x$

a power of $\cos x$

2

nothing

The greatest value of $5 \sin x + 12 \cos x$ is:

12

13

17

5

$\sin x > a$ gives:

points

arcs

one number

nothing

$\tan x \geq 1$ is open at:

$$\frac{\pi}{4}$$

$$\frac{\pi}{2}$$

both ends

neither end

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
General solution	Umumiy yechim	Общее решение
Family of solutions	Yechimlar oilasi	Семейство решений
Homogeneous	Bir jinsli	Однородное
Auxiliary angle	Yordamchi burchak	Вспомогательный угол
Extraneous root	Chet ildiz	Посторонний корень
Diagnosis	Tashxis	Диагностика
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The most common slip in Q1 is:

arithmetic

losing the second family

the period

the sign of π

Q3 requires you to:

reject a root outside $[-1, 1]$

square both sides

divide by $\sin x$

use R

After substituting $t = 2x$ you must:

stop

halve the answers

double them

add π

Lessons 91–98 turn to:

logarithms

probability

vectors

integration

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\sin x = -\frac{\sqrt{3}}{2}$

ANSWER $x = (-1)^{n+1} \frac{\pi}{3} + \pi n$

2. Solve $\cos x = \frac{1}{2}$

ANSWER $x = \pm \frac{\pi}{3} + 2\pi n$

3. Solve $\tan x = -1$

ANSWER $x = -\frac{\pi}{4} + \pi n$

4. R for $\sin x + \sqrt{3} \cos x$

ANSWER 2

5. Reject or keep $t = -2$ in $\sin x = t$

ANSWER Reject

6. Solve $\cos x < \frac{1}{2}$ on $[0, 2\pi]$

ANSWER $(\frac{\pi}{3}, \frac{5\pi}{3})$

7. Period of the tangent

ANSWER π

Medium · the standard question

7 PROBLEMS

1. Solve $\cos 2x = \frac{1}{2}$ on $[0^\circ, 360^\circ]$

ANSWER $30^\circ, 150^\circ, 210^\circ, 330^\circ$

2. Solve $2\sin^2 x + 3 \sin x - 2 = 0$

ANSWER $x = (-1)^n \frac{\pi}{6} + \pi n$

3. Solve $\sin 2x = \sqrt{3} \cos x$

ANSWER $\frac{\pi}{2} + \pi n; (-1)^n \frac{\pi}{3} + \pi n$

4. Solve $\sin x + \sqrt{3} \cos x = 1$

ANSWER $-\frac{\pi}{6} + 2\pi n; \frac{\pi}{2} + 2\pi n$

5. Solve $\sin x \geq \frac{1}{2}$ on $[0, 2\pi]$

ANSWER $[\frac{\pi}{6}, \frac{5\pi}{6}]$

6. Solve $\tan 2x = 1$ on $[0, \pi]$

ANSWER $\frac{\pi}{8}, \frac{5\pi}{8}$

7. Greatest value of $\sin x + \sqrt{3} \cos x$

ANSWER 2

Hard · stretch and reason

7 PROBLEMS

1. Solve $\cos 2x + 3 \sin x = 2$ on $[0, 2\pi]$

ANSWER $\frac{\pi}{6}, \frac{5\pi}{6}$

2. Solve $3\sin^2 x - 4 \sin x \cos x + \cos^2 x = 0$

ANSWER $\frac{\pi}{4} + \pi n; \arctan \frac{1}{3} + \pi n$

3. Solve $\sin 3x = \sin x$ on $[0, \pi]$

ANSWER $0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi$

4. Solve $|\sin x| > \frac{1}{2}$ on $[0, 2\pi]$

ANSWER $(\frac{\pi}{6}, \frac{5\pi}{6}) \cup (\frac{7\pi}{6}, \frac{11\pi}{6})$

5. For which a has $2\sin^2 x - \sin x + a = 0$ a solution?

ANSWER $-3 \leq a \leq \frac{1}{8}$

6. Solve $\sin x + \cos x = \sqrt{2}$ on $[0, 2\pi]$

ANSWER $\frac{\pi}{4}$

7. Solve $\cos^2 x - \sin^2 x > \frac{1}{2}$ on $[0, 2\pi]$

ANSWER $(0, \frac{\pi}{6}) \cup (\frac{5\pi}{6}, \frac{7\pi}{6}) \cup (\frac{11\pi}{6}, 2\pi)$

97 Homework

Homework — 4 tasks

Bring the one-page summary to Lesson 91; probability begins from a blank page.

1. Rewrite in full every control-work question that lost a mark, naming the slip in the margin.

2. Write the four general solutions and the three moves on one page, with a circle drawn for each.
3. Solve $2\cos^2 x - 5 \cos x + 2 = 0$ on $[0, 2\pi]$.
4. Write your target for the probability block in one checkable sentence, and date it.